



Pathology-Aware Speech Encoding and Data Augmentation for Dysarthric Speech Recognition

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Abstract

Automatic speech recognition (ASR) for pathologic speech remains a major challenge due to high variability in articulation, phonation, and prosody distortions. In this work, we propose a pathology-aware speech encoder based on BEST-RQ pre-training, which incorporates 46k hours of speech, including pathologic and atypical speech. We continue pre-training for domain adaptation and experiment with etiology-specific codebooks. We achieve a 13.2% relative word error rate (WER) improvement using the pathology-aware speech encoder with etiology-specific continued pre-training. Additionally, we examine the impact of incorporating synthetic and out-of-domain (OOD) data to further enhance ASR performance. Synthetic data reduces WER by up to 8.7%, while OOD data improves WER by 12.2%. Finally, we introduce a semantic similarity-based data augmentation technique to optimize data selection, achieving a WER improvement of up to 9.7% while minimizing the need for additional training data.

Index Terms: pathologic speech recognition, speech augmentation, self-supervised learning

1. Introduction

Automatic speech recognition (ASR) has made significant progress, yet recognizing pathological speech remains a major challenge due to high variability in articulation, phonation, and prosody distortions. Limited annotated data further exacerbates this issue, making it difficult for ASR models to generalize across different speech impairments. To address these challenges, we introduce a pathology-aware speech encoder trained using BEST-RQ [1] and investigate data augmentation strategies to improve ASR performance. We pre-train a conformer [2] model using BEST-RQ with speech data incorporating both pathologic and healthy speech. We continue pre-training using only pathologic speech and explore etiology-specific codebooks. Additionally, we investigate whether adding more synthetic and OOD data consistently improves ASR performance. Our study also introduces a semantic similarity-based data augmentation approach, which enhances model training by selecting optimal training samples from OOD datasets. By combining these approaches, we aim to improve the robustness and accuracy of ASR systems for pathologic speech. Our findings contribute to the broader field of inclusive speech technology. This work is a contribution to the Speech Accessibility Project Challenge¹.

1.1. Related work

Continued pre-training on domain-specific data has been shown to enhance performance in specialized settings such as medical transcription [3] and accented speech recognition [4]. Hsu et al. [5] found that continued pre-training on in-domain data helps the HuBERT model capture better phonetic representations for low-resource languages. Early approaches to dataset augmentation in ASR primarily involved modifying raw audio signals. Techniques such as time stretching, pitch shifting, and noise addition have been widely adopted to enhance model robustness. Ko et al. [6] introduced SpecAugment, a method that applies time warping, frequency masking, and time masking to spectrogram representations, significantly improving ASR performance without requiring additional labeled data. Another area of research has focused on generating synthetic speech to increase data diversity. Jia et al. [7] demonstrated the effectiveness of text-to-speech (TTS) models in augmenting training datasets, particularly for low-resource languages. Furthermore, Kim et al. [8] explored the use of voice conversion techniques to synthesize speech with varying speaker characteristics, improving ASR generalization. In context of pathologic speech, several works [9, 10, 11, 12, 13, 14] evaluated data augmentation techniques, including input signal modification and generative adversarial networks (GANs) [15].

2. Data

We use the Speech Accessibility Project² 2024-11-30 dataset, which comprises a train and dev set, from 374 speakers and 56 speakers, respectively. In further use, we denote the dataset as SAPC. Speech recordings of five etiologies amyotrophic lateral sclerosis (ALS), cerebral palsy (CP), Down syndrome (DS), Parkinson's Disease (PD) and stroke (SK) are part of the dataset. The most dominant etiology is PD with ~74%, followed by ALS with ~14%, CP with ~6%, DS with ~4% and SK with ~2%. The challenge evaluation is performed on two non-publicly available test sets, test1 and test2, consisting of 7601 and 8043 utterances. We create a separate dev set using 10% of the train speakers for model selection and report the results on the original dev set throughout this work.

Word error rate (WER) and semantic score (SemScore) are the two evaluation measures used in our experiments. The SemScore is a weighted combination of the BERT score [16], phonetic distance³, and natural language inference probability [17]. The metrics used in this work are computed using the provided template by the challenge hosts⁴.

¹<https://eval.ai/web/challenges/challenge-page/2362/overview>

²<https://speechaccessibilityproject.beckman.illinois.edu/>

³<https://ics.uci.edu/~dhirschb/genealogy/Miller/javascrp/soundex.htm>

⁴<https://github.com/xiuwenz2/SAPC-template>

3. Method

3.1. Pathology-aware speech encoder

BERT-based Speech pre-Training with Random-projection Quantizer (BEST-RQ) [1] is a self-supervised learning method for automatic speech recognition (ASR). The core concept involves utilizing a random-projection quantizer (RQ) to convert speech signals into discrete labels, which serve as training targets. The RQ applies a fixed projection matrix to map input features onto a vector aligned with the codebook dimension k . Given an input speech signal X , latent representations are extracted using the convolutional frontend: $H = f_\theta(X)$, where each feature vector h_t lies in a d -dimensional space. To introduce target projection, a fixed, randomly initialized projection matrix $W \in \mathbb{R}^{d \times k}$, where $k < d$, is used to obtain the projected features $Z = HW$. The most similar codebook entry is then identified using cosine similarity, and its index is used as the target label. The RQ operates as a non-trainable transformation that maps high-dimensional speech representations into a lower-dimensional space. Following [18, 19] we employ multiple codebooks during pre-training to stabilize the training. The speech encoder is composed of 12 stacked conformer layers, each incorporating multi-head self-attention with 8 attention heads. Cosine similarity between projected features and codebook entries is used to select the target index. We use cross-entropy and auxiliary Kullback-Leibler divergence loss proposed in [19]. Following [18], we use a multi-codebook setup with 16 equally weighted codebooks in pre-training.

3.1.1. Continued pre-training

To further adapt the pre-trained speech encoder to pathologic speech, we continue pre-training in two ways. Initially, using only SAPC data and equally weighted codebooks following the preceding setup. Subsequently, we experiment with etiology-specific codebooks to determine if speech variations associated with different pathologies can be effectively captured. For this, we employ only the first five codebooks for each etiology and discard the remaining in continued training. During pre-training, the loss for each sample is weighted based on its corresponding etiology. A primary weight w_p is assigned to the codebook specific to the sample’s etiology, while a secondary weight w_s is applied to the remaining codebook losses. To ensure a stronger focus on etiology-specific representations, we enforce the constraint $w_p \geq 6w_s$. This weighting strategy prioritizes learning from pathology-relevant codebooks while still allowing generalization across different etiologies. For both pre-training setups, we use AdamW [20] ($\epsilon = 10^{-9}$, $\beta_1 = 0.9$, $\beta_2 = 0.98$) optimizer and maintain the same learning rate of 2×10^{-4} without warmup, matching the final learning rate used in the multi-corpus pretraining.

3.1.2. Pre-training datasets

We pre-train the speech encoder on SAPC data and expand it with Multilingual LibriSpeech (MLS) [21] (44k hours, predominantly English) for diversity. To enhance robustness, we include 1k hours of children’s speech, as both children and individuals with speech impairments share high acoustic variability, weaker articulation, and altered phonation, resembling characteristics of neurodegenerative conditions (e.g., ALS, PD). We include My Science Tutor [22], CSLU Kids (OGI Kids) [23], PhatSessionz [24], Samrómur Children [25] and corpora from PhonBank: Icelandic Másdóttir [26, 27], Luxembourgish Bracci [28], Portuguese Amorim [29], Portuguese

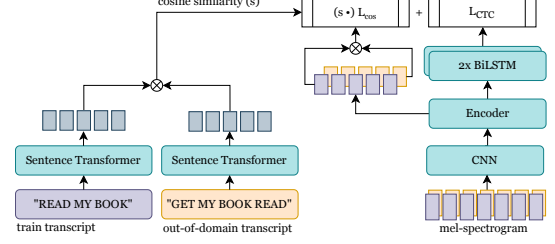


Figure 1: *Semantic similarity augmentation: Out-of-domain utterance selection on the left, fine-tuning using additional OOD utterances and embedding similarity loss on the right.*

Ramalho [30], French Warnier [31], Spanish-Galician Koiné [32], Chilean-Spanish Vivar [33] and Arabic Kuwaiti [34]. All datasets sum up to $\sim 46k$ hours of speech.

3.2. Synthetic data augmentation

We enhance the dataset using XTTS [35] in a zero-shot setting, generating speech from reference audio. Only samples $\geq 15s$ are selected to ensure sufficient reference speech. A time scaling factor $\alpha \sim \mathcal{U}(0.7, 1.0)$ sampled from a uniform distribution was used to simulate the slower speech rates occurring in pathologic speech. Transcripts of the SAPC train set are employed as prompt for generation.

3.3. Out-of-domain data augmentation

Unlike synthetic data, which demands computational resources even for zero-shot generation, we expand the fine-tuning dataset beyond SAPC data by incorporating OOD datasets. Specifically, we utilize widely employed datasets such as LibriSpeech [36], Common Voice [37], and People’s Speech [38]. For our experiments, we select an equal number of samples from each dataset, a maximum of $\sim 66k$ samples of each dataset. Additionally, we investigate the impact of OOD data on fine-tuning by using varying portions of LibriSpeech, Common Voice, and People’s Speech.

3.3.1. Semantic similarity data augmentation

Instead of randomly selecting OOD speech samples, we employ a semantic similarity-based approach using Sentence Transformer [39, 40] embeddings. For each unique transcript in the SAPC dataset, we identify the most semantically similar and the most dissimilar utterances from the OOD datasets. The selection process relies on cosine similarity scores computed between the embedding representations of the transcripts. During fine-tuning, we construct training triplets by selecting the closest and farthest OOD utterances: (1) the SAPC utterance as the anchor $\mathbf{a}$, (2) the most similar OOD utterance as the positive $\mathbf{p}$, and (3) the most dissimilar OOD utterance as the negative $\mathbf{n}$. The additional loss is calculated on the mean embeddings from the speech encoder and can be formulated as:

$$\mathcal{L}_{\text{sem}} = (1 - \cos(\mathbf{a}, \mathbf{p})) + (1 + \cos(\mathbf{a}, \mathbf{n})) \quad (1)$$

Since not every transcript in the SAPC dataset has a closely matching OOD utterance, we introduce an additional weight based on the cosine similarity of the transcript embeddings. This way, not appropriately matching OOD utterances are less weighted. The overall proposed setup is illustrated in Figure 1.

4. Experiments

All experiments share the same fine-tuning setup. Fine-tuning is performed by adding a two layer bidirectional LSTM [41, 42] on top of the encoder. The learning rate is set to 3×10^{-4} , AdamW ($\epsilon = 10^{-9}$, $\beta_1 = 0.9$, $\beta_2 = 0.98$) is used as optimizer. We employ connectionist temporal classification (CTC) [43] objective and fine-tune all models for 30 epochs. Best model is selected based on the WER of our dev set.

4.1. Baselines

We begin by utilizing widely adopted pre-trained speech encoders, including wav2vec 2.0 [44], HuBERT [5], and BEST-RQ [19], which we fine-tune on the SAPC dataset to establish our baselines. All models are used in their self-supervised pre-trained versions, ensuring direct comparability as they fall within a similar parameter range. Table 1 presents the baseline results. Among them, wav2vec 2.0 performs the worst overall, while BEST-RQ achieves the best fine-tuning results, with a WER of 22.73 and a SemScore of 83.01. However, a more detailed analysis across different etiologies reveals that while BEST-RQ is superior for PD, HuBERT outperforms the other models across all remaining etiologies.

4.2. Pathology-aware speech encoder

Building on the baseline experiments, we pre-train a pathology-aware speech encoder using BEST-RQ. We choose BEST-RQ for its simpler training process, avoiding the multi-stage pre-training and clustering required by HuBERT, while ensuring greater stability than wav2vec 2.0. We first pre-train the speech encoder on SAPC data, assigning a separate codebook per etiology, following the feature-level clustering approach in [19]. As shown in Table 2, this pre-training approach leads to lower WER and SemScore compared to the same model pre-trained on LibriSpeech in Table 1. To enhance performance, we pre-train the speech encoder on a diverse 46k-hour dataset, described in Section 3.1.2. The results in Table 2 show that the multi-corpus pre-trained speech encoder (base-MC) achieves an 11.8% relative WER improvement over the baseline model.

4.2.1. Effect of continued pre-training

To examine continued pre-training for domain adaptation, we refine the 46k-hour pre-trained model by further training on SAPC data for 20k additional steps. We compare two setups: (1) shared codebooks, following the initial pre-training, and (2) etiology-specific codebooks for targeted adaptation. Table 2 presents the results for both, continued pre-training and etiology-specific codebook pre-training. Continued pre-training with shared codebooks does not show a significant positive or negative effect on the general performance metrics. However, incorporating etiology-specific codebooks during pre-training yields the best WER and SemScore results compared to the shared codebooks setup. Using etiology-specific codebooks in continued pre-training (base-ESC) not only improves overall performance — achieving a 13.2% relative WER reduction — but also leads to notable improvements for the ALS and CP etiologies. Given its superior performance, we select the base-ESC model for subsequent experiments.

4.3. Synthetic data augmentation

We expand the dataset by generating XTTS synthetic speech using SAPC reference audio, at 50% to 100% of the training

set. While XTTS improves WER, it is computationally intensive. As shown in Table 3, 50% synthetic augmentation reduces WER by 8.7%, benefiting underrepresented etiologies (ALS, CP). However, exceeding 100% synthetic data leads to diminishing returns, likely due to artifacts affecting generalization.

4.4. Out-of-domain augmentation

We evaluate OOD data from LibriSpeech, Common Voice, and People’s Speech for fine-tuning, optimizing augmentation ratios for ASR performance. As shown in Table 3, OOD data consistently improves generalization, with the best results at 150% augmentation, reducing WER to 17.32% and increasing SemScore to 86.12. However, excessive OOD data slightly degrades performance, especially for DS, emphasizing the need for careful data selection to match pathological speech characteristics.

4.5. Semantic similarity augmentation

To identify the most similar and dissimilar utterances from OOD datasets, we leverage Sentence Transformer (ST) embeddings, extracted from transcripts. The selected utterances account for an additional $\sim 29\%$ of the SAPC dataset. For embedding extraction, we use the pre-trained `all-MiniLM-L6-v2` model across all transcripts. We begin with a naïve approach, where each SAPC utterance is paired with its most similar OOD counterpart in training batches (OOD-b) without further modifications. Next, we introduce an additional cosine similarity loss between the embeddings of a SAPC utterance and its most similar OOD utterance (OOD-bl). This loss is further weighted by the similarity score computed from the ST embeddings (OOD-blw). Inspired by triplet loss, we extend the approach by incorporating the most dissimilar utterance alongside the most similar one (OOD-t). Finally, we apply per triplet of samples a loss weighted by the transcript similarity score (OOD-tw).

Table 3 presents the results of fine-tuning with semantically selected utterances from OOD datasets. The naïve approach without an additional loss yields diminishing returns. Introducing an embedding loss (OOD-bl) between only positive pairs results in a 9.1% relative WER improvement. Weighting the loss by transcript cosine similarities (OOD-blw) leads to a slight but consistent WER improvement, from 17.93% to 17.81%, across almost all etiologies. In contrast, using unweighted triplets with an additional embedding loss (OOD-t) performs worse than positive pairs alone, and applying weighting (OOD-tw) does not yield significant gains. The training process suggests that the harder training objective may require longer fine-tuning to surpass the performance of positive pairs alone.

5. Discussion

This study demonstrates that pre-training a speech encoder on diverse datasets significantly improves ASR performance for pathological speech. By leveraging a pathology-aware speech encoder with BEST-RQ and incorporating multiple data augmentation strategies, we effectively mitigate the challenges posed by high speech variability and data scarcity.

Our findings highlight the effectiveness of etiology-specific codebooks in capturing distinct speech patterns associated with different pathologies. This approach led to a 13.2% relative WER improvement compared to baseline models, confirming that targeted representation learning enhances ASR accuracy for dysarthric speech. While continued pre-training using SAPC alone did not significantly impact performance, the integration of etiology-specific codebooks proved beneficial, particularly

Table 1: Baseline models WER ($\downarrow$) and SemScore ($\uparrow$) per etiology. Each model is pre-trained on 960 hours of speech from LibriSpeech. Underlined results are the best per table, while bold results indicate overall best results.

Model	ALS		CP		DS		PD		SK		Overall	
	WER	SemScore	WER	SemScore	WER	SemScore	WER	SemScore	WER	SemScore	WER	SemScore
wav2vec 2.0	18.86	84.75	32.95	72.61	30.06	72.91	24.49	83.34	12.95	88.90	24.51	82.18
HuBERT	<u>18.34</u>	<u>84.82</u>	<u>31.73</u>	<u>73.87</u>	<u>28.53</u>	74.80	23.98	84.01	<u>12.89</u>	<u>89.27</u>	23.84	82.91
BEST-RQ	18.65	84.25	32.87	72.20	33.06	71.49	<u>22.19</u>	<u>84.81</u>	13.47	88.31	<u>22.73</u>	<u>83.01</u>

Table 2: BEST-RQ conformer WER and SemScore per etiology for SAPC only pre-training (base-SAPC), multi-corpus pre-training (base-MC), continued pre-training (base-C) using SAPC data and continued etiology-specific codebook pre-training (base-ESC).

Model	ALS		CP		DS		PD		SK		Overall	
	WER	SemScore	WER	SemScore	WER	SemScore	WER	SemScore	WER	SemScore	WER	SemScore
base-SAPC	26.28	80.32	40.13	67.18	37.14	67.71	31.47	80.38	15.01	87.19	31.44	78.84
base-MC	16.17	85.14	29.92	73.61	29.25	<u>74.09</u>	19.47	86.05	<u>10.05</u>	90.53	20.04	84.50
base-C	15.86	85.27	30.05	73.43	29.84	73.31	19.60	85.91	10.77	90.10	20.00	84.34
base-ESC	<u>15.53</u>	<u>85.52</u>	<u>29.06</u>	<u>74.52</u>	28.51	74.04	<u>19.26</u>	<u>86.39</u>	10.09	<u>90.59</u>	<u>19.73</u>	<u>84.78</u>

Table 3: Synthetic and out-of-domain data augmentation in fine-tuning using the BEST-RQ pretrained conformer with continued pre-training using etiology-specific codebooks (ESC).

Model	ALS		CP		DS		PD		SK		Overall	
	WER	SemScore	WER	SemScore	WER	SemScore	WER	SemScore	WER	SemScore	WER	SemScore
base-ESC	15.53	85.52	29.06	74.52	28.51	74.04	19.26	86.39	10.09	90.59	19.73	84.78
XTTS-50	13.83	85.97	27.95	74.74	29.73	72.02	17.48	87.85	10.47	91.03	18.02	85.83
XTTS-100	13.77	85.97	28.23	74.72	30.89	71.11	17.56	87.67	10.62	90.75	18.13	85.61
OOD-50	14.42	86.07	28.77	74.68	31.45	70.29	17.52	88.15	10.47	90.58	18.20	85.97
OOD-100	13.93	86.43	28.46	74.55	31.96	70.28	16.97	88.14	9.84	90.86	17.71	85.90
OOD-150	13.33	86.70	28.23	74.20	32.30	69.96	16.57	88.39	9.84	91.47	17.32	86.12
OOD-b	16.34	84.72	32.18	72.55	31.33	70.91	19.25	86.37	9.92	90.63	19.92	84.76
OOD-bl	14.31	86.22	28.49	74.73	31.27	70.89	17.24	89.01	9.87	90.84	17.93	86.47
OOD-blw	14.25	86.20	28.32	74.91	31.35	70.76	17.13	89.27	9.81	90.88	17.81	86.52
OOD-t	14.54	85.97	28.62	<u>74.61</u>	31.47	70.29	17.53	<u>88.58</u>	<u>10.14</u>	90.16	18.12	<u>86.07</u>
OOD-tw	14.50	85.80	28.56	74.73	31.44	70.21	17.46	88.69	10.23	89.91	18.05	86.19

for ALS and CP speech. These results suggest that incorporating disorder-specific representations helps models better generalize.

Data augmentation emerged as a crucial factor in improving ASR generalization, with both synthetic and OOD data playing a key role. Using XTTS-generated speech provided an 8.7% relative WER reduction. However, our results indicate diminishing returns when the synthetic-to-real data ratio exceeds 100%, likely due to the introduction of artifacts that affect model generalization. Incorporating speech from external datasets such as LibriSpeech, Common Voice, and People’s Speech improved WER by 12.2% at an optimal augmentation ratio of 150%. Selecting OOD utterances based on Sentence Transformer embeddings further refined augmentation by prioritizing semantically similar data. This method resulted in a 9.7% relative WER improvement, demonstrating that quality selection of augmentation data is more impactful than simply increasing dataset size.

Despite these advancements, our study identified areas that require further investigation. Unlike other etiologies, DS speech did not benefit from augmentation techniques at all. Only the etiology-specific pre-training improved the results. Future research could analyze phonetic and acoustic properties of DS

speech to develop specialized augmentation strategies. The impact of extended fine-tuning on harder training objectives, such as weighted triplet loss, remains unexplored and requires further research.

6. Conclusion

We introduced a pathology-aware speech encoder utilizing BEST-RQ pre-training and evaluated multiple data augmentation techniques to improve ASR performance for pathologic speech. Our results demonstrate that etiology-specific codebooks effectively enhance recognition accuracy, yielding an overall 13.2% improvement in WER compared to baseline. This underscores the importance of explicitly modeling pathology-specific variations in speech processing. Additionally, we found that synthetic data augmentation significantly reduces WER, particularly when used in moderate amounts, while excessive reliance on synthetic data can lead to diminishing returns. Out-of-domain data augmentation further improved ASR generalization, with the best performance observed with most out-of-domain data. The semantic similarity-based augmentation strategy also contributed positively, demonstrating that targeted transcript selection enhances fine-tuning efficiency.

7. References

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